

Preliminary analysis of selected status files for wind turbine SCADA datasets

Kelmarsh

We have looked into first dataset that consisted of six wind turbines from Kelmarsh wind farm located in the UK. Status information for wind turbines corresponds with those that we have taken account into during general analysis. Data was collected from 1st of January to 30th of June 2021. Timestamps are more irregular, since this dataset contains information about error and status codes that might occur more randomly.

Results

The following results were obtained during the first analysis of the data:

	Turbine 01	Turbine 02	Turbine 03	Turbine 04	Turbine 05	Turbine 06
Columns (variables)	9	9	9	9	9	9
Datapoints	28565	21390	32399	30871	30095	37159
First timestamp	2020-12-31 04:43:55	2020-12-31 06:14:21	2021-01-01 09:47:28	2020-12-28 09:23:24	2020-12-31 06:02:04	2020-06-07 12:20:51
Last timestamp	2021-06-30 23:50:37	2021-06-30 18:28:41	2021-06-30 19:12:49	2021-06-30 19:21:45	2021-06-30 22:54:51	2021-06-30 23:17:34
Missing values	17191	13359	20341	18584	18145	22421

We have noticed that number of missing values is significant. After taking a closer look into the data we have noticed some correlations between data points.

- 1. **Timestamps and duration**
There were 3 values that described position in time for the windturbine. Timestamp start, timestamp end and duration. Timestamp start is never missing, but duration is tightly coupled to Timestamp end. If we have Timestamp end we also have duration, thus in reality we have less missing values.
- 2. **Status, codes and messages**
Those described type, code and message of a status of a turbine and were never missing.
- 3. **Comment**
This row was always empty, so we do not have to pay attention to it.
- 4. **Service contract category and IEC category**
Number of missing service contract category is significant, but IEC category is almost always present.

Real number of missing values

Since we have eliminated some missing values from the analysis, we could calculate real number of missing values:

	Turbine 01	Turbine 02	Turbine 03	Turbine 04	Turbine 05	Turbine 06
Datapoints	28565	21390	32399	30871	30095	37159
Missing values	17191	13359	20341	18584	18145	22421
Missing values without irrelevant data	7881	6292	9588	8540	8322	10398

We can see that this number is halved compared to previous result.

Penmanshiel

Situation for this British wind farm located in Penmanshiel is similar to Kelmarsh. This time 5 different turbines were taken into account. Irrelevant or redundant data was also calculated to present following result:

Results

	Turbine 01	Turbine 02	Turbine 03	Turbine 04	Turbine 05
Columns (variables)	9	9	9	9	9
Datapoints	31352	26157	32117	31486	26687

First timestamp	2021-01-01 Turbine 01 09:47:45	2021-01-01 Turbine 02 09:47:44	2021-01-01 Turbine 03 09:47:51	2021-01-01 Turbine 04 09:47:51	2020-12-21 Turbine 05 17:08:09
Last timestamp	2021-06-30 23:06:10	2021-06-30 22:36:29	2021-06-30 22:35:17	2021-06-30 22:41:56	2021-06-30 22:41:15
Missing values	19633	16350	19651	19751	16738
Missing values without irrelevant data	9099	7618	9105	9148	7776

We can see that number of missing values halved after eliminating coupled and irrelevant data.